

**Chinese Naval Access and Military Infrastructure Expansion at Ream Naval Base,
Cambodia**

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Executive Summary

This project evaluates the modernization and expansion of Ream Naval Base in Preah Sihanouk Province, Cambodia, and looks into the likelihood that the facility is being developed to support access by the China's People's Liberation Army Navy (PLAN). The analysis consists of commercial and open-source material. Imagery analysis is mainly taken from Copernicus Sentinel-2 and SAR coverage with Google and Apple Maps basemaps for high-resolution baseline imagery. Public reporting is drawn from the CSIS Asia Maritime Transparency Initiative (AMTI), Radio Free Asia, Reuters, Nikkei Asia, and the Lowy Institute.

Ream Naval Base has very likely been developed into a PLAN forward access and logistics center on the Gulf of Thailand. The base lies on Cambodia's only stretch of coastline on that gulf, between Thailand and southern Vietnam, near both the South China Sea and the western approaches to the Strait of Malacca, a strategically valuable location. U.S.- and Australian-funded facilities at the base were demolished in 2020 after which Cambodia and China launched a Chinese-funded modernization project, with an official groundbreaking in mid-2022. Since then, open-source imagery and reporting point towards a rapid build-out: a deep-draft pier capable of berthing vessels much bigger than anything in the Royal Cambodian Navy's (RCN) inventory, a new dry dock and ship ramp, an extended wharf, dredged approach channels, headquarters and support buildings, earth-covered ammunition magazines, and a China-Cambodia joint support and training center that formally opened in April 2025. It is almost certain that this physical build-out exceeds RCN's own requirements.

The clearest sign is the use of the new pier. Since December 2023, two Chinese PLAN Type-056A corvettes have stayed at the new deep-draft pier, rotating in and out over time. Cambodian ships have continued using the older southern pier, and no other foreign navy's warships have been seen using the new one. This pattern suggests China has privileged, and possibly exclusive, access to the pier. This does not match Cambodia's public claim that Ream is an open port for multiple navies. Ream also fits a wider pattern of Chinese overseas access because like Gwadar, Hambantota, and Bata, it appears to give China useful military access without officially calling the site a permanent base. The facility can support a small but continuous Chinese presence, provide logistics access, and make it easier for China to monitor activity in the region. The pattern has not been completely one-sided. In January 2026 there was a

new development with the USS Cincinnati becoming the first U.S. warship to dock at Ream since the Chinese-funded renovation began. This happened during a visit by the Commander of U.S. Indo-Pacific Command, along with announcements that the Angkor Sentinel exercise would restart and that the U.S. arms embargo on Cambodia would be lifted. It is possible to see this as Cambodia signaling that it still wants some balance in its foreign relationships and not as proof that Chinese access has ended. One planned U.S. visit does not erase nearly two years of regular Chinese naval presence. The physical expansion of Ream and the way the pier has been used make it very likely that the base is being developed into a regular PLAN access point. However, open sources still cannot confirm the exact terms of any Cambodia-China agreement.

Key Judgments

- It is highly likely that Ream has undergone major Chinese-funded expansion adding deep-draft berthing, dry-dock repair, and support infrastructure exceeding the requirements of the current RCN fleet.
- China likely holds privileged access to the new deep-draft pier, based on continued PLAN corvette presence and the absence of any confirmed foreign warship berthing at that pier. A January 2026 U.S. Navy port call is assessed as a Cambodian hedging signal that does not, by itself, overturn this pattern.
- Cambodian information-management around the northern half of the base with a near-total absence of official imagery from that sector, the 2021 denial of base access to a U.S. defense attaché, and the demolition of the former U.S.- and Australian-funded facilities are consistent with the deliberate concealment of privileged foreign use.
- Ream is likely being developed into a PLAN forward logistics and presence node
- If privileged access continues, the base would expand PLAN persistence and power-projection toward the Strait of Malacca approaches and the southwestern part of the South China Sea

Methodology and Framework

This assessment's approach uses multi-source change detection. A pre-construction 2020 baseline was used and tracked forward using the optical scenes on Copernicus Sentinel-2 (~10 m), Sentinel-1 C-band SAR for cloud-obscured intervals common to the Gulf of Thailand coast, the SOAR Atlas archive, and Google and Apple Maps basemaps. Open-source reporting including the CSIS Asia Maritime Transparency Initiative (AMTI), Radio Free Asia, Reuters, Nikkei Asia, and the Lowy Institute was used to figure out the chronology of events and corroborate change. Independent structural measurements were taken from available imagery.

Research organizations conducting their own imagery analysis or in-country fieldwork (AMTI, Lowy) are treated as higher-reliability than single-outlet press accounts, and statements from parties that might have an interest (Phnom Penh, Beijing) are treated as claims to be tested rather than as established fact. Where a judgment rests on a single line of reasoning,, most notably the dry-dock user-class hypothesis, this is stated explicitly and the competing explanation is retained. A structured comparison of the principal competing hypotheses is provided in the Alternative Hypotheses section.

Imagery Analysis



Figure 1: High-resolution overview of Ream Naval Base showing the Chinese-funded deep-draft pier to the north and the original Cambodian (RCN) pier to the south. (Apple Maps / commercial imagery, 2025)

Ream Naval Base is located on the Gulf of Thailand coast of Preah Sihanouk Province (10.51° N, 103.62° E.) It is about 30 km southeast of Sihanoukville city. The Chinese-funded deep-draft pier takes up the southwestern waterfront while the original Royal Cambodian Navy pier lies immediately to its south. (10.5122°N, 103.6046°E, MGRS 48P US 47302 62383.)



Figure 2: Copernicus Sentinel-2 coverage of the base. At 10 m resolution the pier and reclaimed waterfront are identifiable but individual structures and vessels cannot be resolved. (Copernicus Sentinel-2, 2025)

The baseline period shows demolition of the older western facilities in 2020, followed by land clearing and the first new construction from 2022 onward. The clearest visible changes are on the northern half of the base, which is the section most widely reported as being set aside for Chinese use. There are also clear changes along the southwestern waterfront, where a new pier, wharf, dry dock, and ramp have taken shape.



Figure 3: Mid-construction imagery showing land clearing and new building rows (Google Earth, 2023)

SAR coverage is useful in this situation because it penetrates the persistent cloud cover and bright radar over the new pier and reclaimed ground help confirm hard infrastructure rather than just transient activity. Dredging signs along the waterfront also line up with deepening the approaches to accommodate deeper-draft ships. The pace of construction is also an indicator. Reporting based on higher-resolution commercial imagery describes multiple large tracts cleared and buildings being completed within weeks. It is also worth noting that there was a finger pier that outside analysts have noted closely resembles the PLAN pier at China's Djibouti base. Figure 4 presents Sentinel-1 imagery over the waterfront. The strong stable backscatter over the pier and reclaimed apron is consistent with fixed metal-and-concrete infrastructure and not transient surface activity. This points towards the construction footprint being during an interval when optical coverage was cloud-obscured. SAR is used to give an independent material-return check on hard infrastructure.

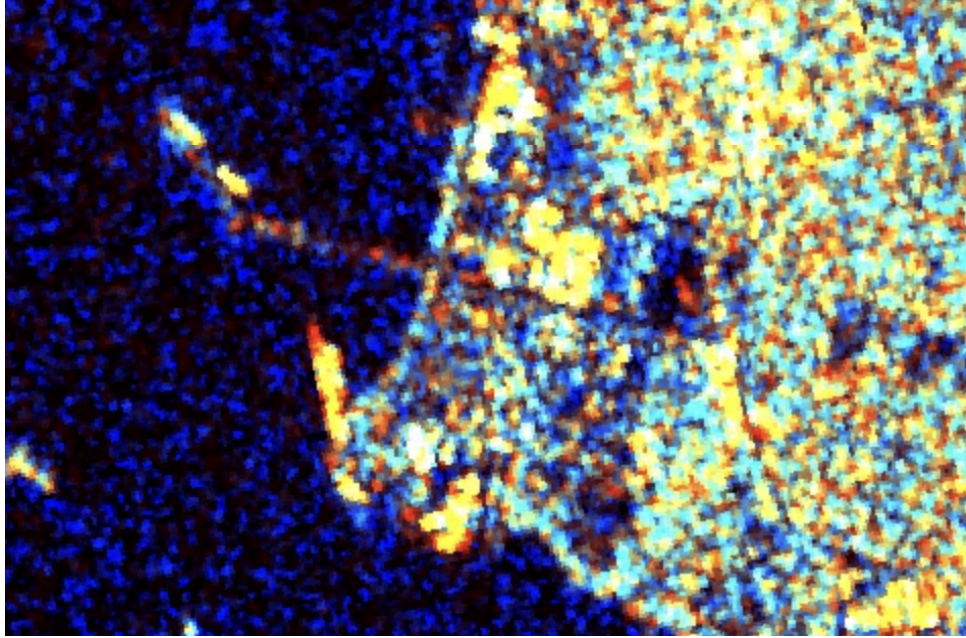


Figure 4: Sentinel-1 C-band SAR scene of the Ream waterfront. Persistent bright returns over the deep-draft pier and reclaimed ground confirm stable hard infrastructure; the radar backscatter penetrates the cloud cover that creates gaps in the optical timeline. (Copernicus Sentinel-1, 2023)



Figure 5: Active construction in the northern half of the base barracks-style buildings, support structures, and graded roadways consistent with the reported rapid build-out. (Commercial imagery, 2023)

Vessel-presence tracking by AMTI found the two PLAN corvettes berthed at the new pier for around 93 percent of the first months of 2024, with only a couple short vacancies. The corvettes have rotated at least twice since then through 2025 and into 2026.

The vessels stationed there are assessed as being Jiangdao-class PLAN Type-056A corvettes. Two have maintained the near continuous presence, and no vessel larger than a corvette has been seen berthed at the deep draft pier in available coverage which is notable given that the pier is sized for far larger ships. There is some open reporting characterizing the two corvettes as transfers or "gifts" to the Royal Cambodian Navy rather than as deployed PLAN units. This distinction is consequential. If the hulls are Cambodian-operated, their presence is less of a confirmation of privileged PLAN access; if they remain PLAN-crewed, the presence is likely confirmed. AMTI has tracked them as a sustained PLAN-pattern presence but their operational status is unconfirmed in open sources so it is treated as an open question in this project

Assessment: It is almost certain that the imagery record reflects a deliberate, externally resourced expansion rather than incremental Cambodian upgrades, and very likely that the new deep-draft berth has been used on a near-continuous basis by PLAN vessels. High confidence on the build-out; moderate-to-high on the access pattern.

Limitations: Sentinel-2 resolution does not support reliable identification of individual hulls, weapons, or sensors; those details rest on secondary reporting using commercial imagery whose dates do not always align with the events described. Cloud cover and revisit gaps leave intervals where change can be bounded only approximately.

Technical Factors

Several fixed-infrastructure indicators distinguish a modest coastal patrol base from a facility built to host and sustain larger or foreign warships. Each is assessed against the RCN's known requirements, which center on small patrol craft.

Berthing Capacity



Figure 6: The Chinese-funded deep-draft pier with two PLAN-pattern corvettes berthed. The pier was independently measured at approximately 340 m — close to the lower reported figure and far beyond Royal Cambodian Navy berthing needs. (Commercial imagery, 2025)

The new deep-draft pier is the most significant indicator. Public reporting describes it in multiple different ways. Figures around 363 meters and around 650 meters come up in different accounts, and it is characterized as capable of berthing vessels up to and including aircraft carriers, which the RCN does not operate and has no announced plans to acquire. Measurement of the imagery placed the pier at approximately 340 m, close to the lower reported figure. The discrepancy in reported length is itself an intelligence gap, but in either case the berth is sized beyond Cambodian fleet needs. A pier built for ships a navy does not own is a strong indicator that the builder anticipates use by another navy's vessels.

The pier measurement was taken by scaling the two berthed Type-056A corvettes against a known-dimension reference feature in the available oblique imagery; given the viewing angle and resolution, the figure is assessed as accurate to within roughly 15 to 20 m. Part of the wide discrepancy in reported lengths likely reflects different sources measuring different features; the deep-draft finger pier, the separate wharf, or an end to end span including the connecting causeway, rather than disagreement about a single structure. The 340 m measurement is most consistent with the finger pier alone and favors the lower reported figure ~363 m over the 650 m account. Imagery also shows the original smaller pier in continued use by Cambodian small combatants



Figure 7: The original Cambodian pier with small combatants berthed, consistent with continued Royal Cambodian Navy use of the older facility while the new deep-draft pier is reserved for PLAN vessels. (Commercial imagery, 2025)



Figure 8: The new dry dock (measured at ~133 m) with crane and marine railway/ramp (~230 m). The drydock-plus-ramp configuration resembles the layout at the PLAN submarine base at Qingdao. (Commercial imagery, 2025)

The new dry dock and adjacent ramp/slipway add a maintenance and repair capability the base previously lacked (Figure 8). One widely cited analysis judges the dry dock, reported at roughly 140 meters, and measured here at approximately 133 m which is too short for the PLAN's larger surface combatants but appropriately sized for submarines or small combatants, and notes a layout resembling the drydock-plus-ramp arrangement at the PLAN submarine base at Qingdao. This is a plausible but not confirmed reading: the same facility could be intended primarily to service the resident corvettes and RCN craft. A further consideration weighing against the submarine reading is that the Gulf of Thailand is comparatively shallow, which some analysts argue would limit covert submarine operations from Ream. I therefore hold the submarine-optimization line at low-to-moderate confidence and treat it as a hypothesis warranting collection rather than a finding. The dry-dock measurement (~133 m) is similarly approximate (± 10 m) and is consistent with the around 140 m figure in secondary reporting.

Support, Command, and Ordnance Infrastructure



Figure 9: Radome at Ream, consistent with reporting of a naval radar/sensor capability. (Commercial imagery, 2025)



Figure 10: New headquarters/administrative building under construction in the base interior, consistent with reporting of work begun in late 2024. (Commercial imagery, 2025)



Figure 11: Multi-story barracks and support buildings with an adjacent helipad, indicating expanded garrison and rotary-wing support capacity. (Commercial imagery, 2025)

Imagery and reporting also document earth-covered ammunition magazines at the northern tip of the base, new headquarters and administrative buildings begun in late 2024 (Figure 10), dredged navigation channels, a radome consistent with a naval radar capability (Figure 9), and the China-Cambodia joint support and training center opened in April 2025. Expanded barracks and support housing with an adjacent helipad are also visible (Figure 11). Separately, a development site roughly half a mile north of the base, reported as planned for an air-defense command headquarters and a naval radar system; saw clearing and preparation through 2024 before activity appeared to stall in available imagery. Magazines, fixed command facilities, and radar infrastructure are consistent with sustained operational use rather than occasional port calls. Assessment: The combination of carrier-capable berthing, a dedicated repair dock, ordnance storage, and command and training facilities makes it almost certain that Ream is being equipped for sustained naval operations beyond the RCN's organic capacity. (High confidence.) The specific intended user-class of the dry dock is unresolved. (Low-to-moderate confidence.)

Limitations: Independent measurements were approximate, constrained by oblique viewing angle and resolution, and the pier-length discrepancy is not fully resolved. The presence of weapons, sensors, or fitted-out interiors cannot be confirmed from open optical imagery.

Access, Concealment, and Denial Indicators

Behavioral and access-control indicators independently support the privileged-access judgment. In 2021, Cambodia denied the U.S. defense attaché full access to the base during an invited visit, and the former U.S.- and Australian-funded facilities were demolished beginning in 2020 ahead of the Chinese-funded rebuild. Most tellingly, an open-source review by AMTI of the Cambodian military's own extensive photographic record of activity at Ream found that, across thousands of images of ceremonies, maintenance, and official visits over the preceding year, virtually none were taken on the northern half of the base, the sector most widely reported as set aside for Chinese use, apart from the April 2025 opening ceremony.

A consistent absence of imagery from a specific sector of an otherwise heavily documented base is itself an information-management indicator. Taken together, restricted physical access, the displacement of the prior U.S. presence, and the selective documentation gap are more consistent with deliberate concealment of privileged foreign use than with the open, multinational role Cambodian officials describe. (Moderate confidence.)

Regional Power Projection

Cambodia's Strategic Calculus

The durability of any access arrangement depends as much on Cambodian motives as on Chinese activity. Open reporting and regional analysis indicate that Phnom Penh's pursuit of Chinese-financed modernization is driven less by alignment than by hedging: Cambodia seeks an external patron to modernize a chronically underfunded military against perceived security pressures from Thailand and Vietnam, while accepting Chinese financing and influence as the cost. Heavy economic dependence on Chinese investment and lending, and the consolidation of power across the 2023 leadership transition from Hun Sen to Hun Manet, both reinforce the

relationship. Cambodia has also historically been willing to act on Beijing's behalf within ASEAN, including blocking consensus language on the South China Sea.

The hedging interpretation is reinforced by the January 2026 reopening to U.S. visits: a state pursuing pure alignment with China would have less incentive to stage a high-profile American port call. This implies the arrangement is politically contingent and potentially reversible, an important caveat to the basing judgment.

Basing Pattern and Geography

Ream fits an observable pattern in Chinese overseas access development. Rather than declared bases on the Djibouti model, which carries diplomatic cost and scrutiny, China has cultivated a network of dual-use facilities, including Gwadar in Pakistan, Hambantota in Sri Lanka, and Bata in Equatorial Guinea, that afford preferential access, a small persistent footprint, and intelligence-collection geometry without a formal basing agreement. The Ream finger pier's reported resemblance to the PLAN pier at Djibouti, and the drydock plus ramp layout's resemblance to the PLAN submarine base at Qingdao, are consistent with this pattern. Read against the Djibouti precedent, the sequence at Ream, pier, repair capability, ordnance storage, command and training facilities, sustained vessel presence, tracks the early stages of a persistent-access node rather than a one-off upgrade.

If China retains privileged access, Ream would give the PLAN its first persistent foothold on the Gulf of Thailand and its second acknowledged overseas access point after Djibouti. Geographically, the base flanks the South China Sea from the southwest and sits markedly closer to the Strait of Malacca than China's submarine base on Hainan, a distance that matters most for slower conventional submarines and for the resupply and presence missions that underpin China's stated interest in protecting maritime supply routes through the Malacca chokepoint. Measured from Singapore, Ream lies roughly 660 miles away versus about 1,340 miles for Hainan; analysts differ on whether this roughly 700-mile difference is strategically decisive or only a marginal improvement.

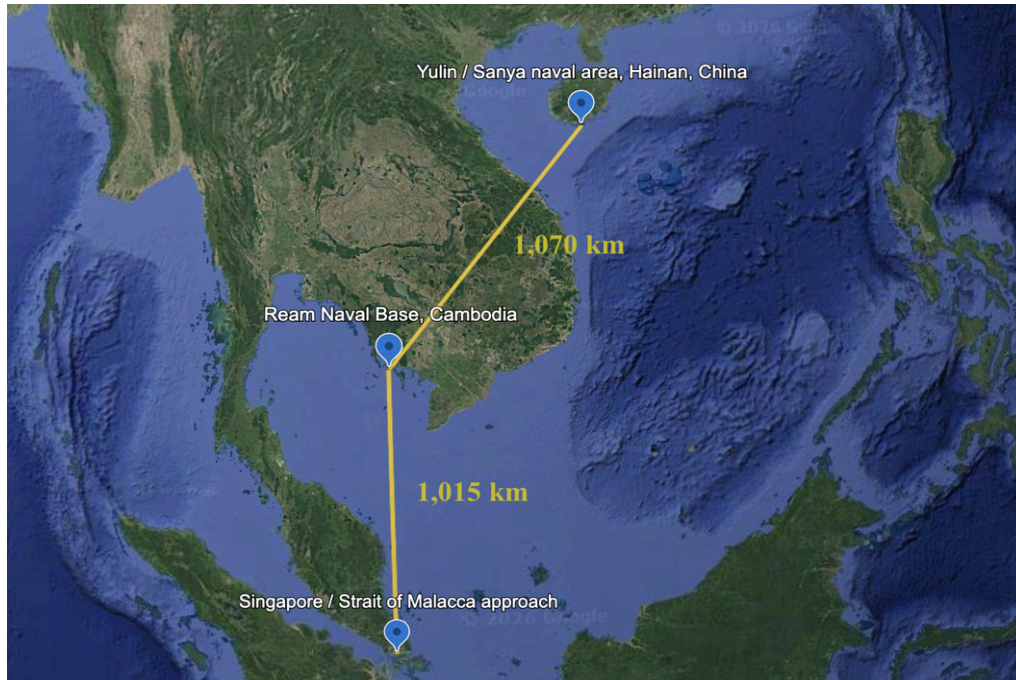


Figure 12: Comparative distance to the Strait of Malacca. Arcs measured from Singapore show Ream at roughly 660 mi versus approximately 1,340 mi for China's submarine base on Hainan, illustrating the ~700 mi reduction in transit distance to the Malacca approaches. (Author graphic over commercial basemap, 2025)

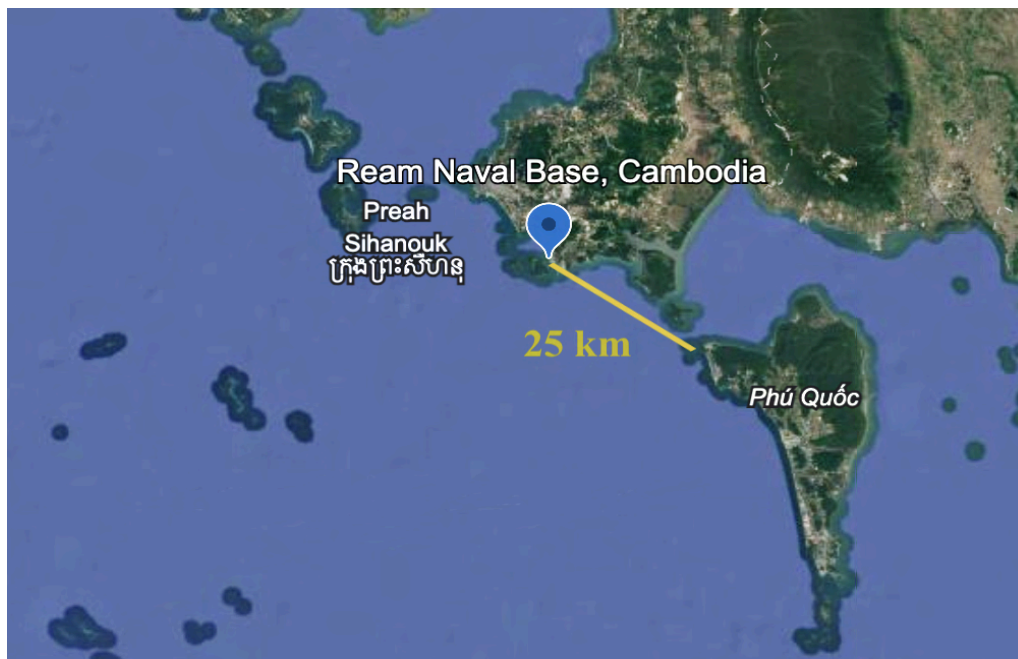


Figure 13: Regional maritime setting. Ream sits on the Gulf of Thailand directly across from Vietnam's Phu Quoc island, less than 30 km away, underscoring the base's proximity to Vietnamese territory. (Commercial satellite basemap, 2025)

Operationally, persistent access would extend PLAN dwell time, repair reach, and intelligence-collection geometry in the lower South China Sea and Gulf of Thailand without requiring a declared permanent base. It would also complicate the posture of the United States and partners such as Japan and Vietnam: the sustained PLAN occupancy of the new pier, alongside the absence of any confirmed foreign warship at that berth, suggests the facility is not, in practice, the open multinational port Phnom Penh describes. For Vietnam in particular, a PLAN-accessible base on its southwestern maritime flank is a notable proximity concern, with Ream lying less than 30 km from Vietnam's Phu Quoc island.

This reading was tested in January 2026, when the USS Cincinnati, a San Diego-based Independence-variant littoral combat ship (around 130 m, around 3,000 tons), became the first U.S. Navy warship to dock at Ream since the Chinese-funded renovation, coinciding with a visit by the Commander of U.S. Indo-Pacific Command and announcements that the Angkor Sentinel exercise would resume and a U.S. arms embargo would be lifted. The Cincinnati was the third foreign naval vessel to call at the modernized base, following earlier Japanese visits. The visit demonstrates that Ream is not formally closed to foreign navies and complicates any claim of strict exclusivity. It does not, however, overturn the privileged-access assessment, for three reasons: a single, pre-announced, choreographed confidence-building call is categorically different from a two-year sustained operational presence; Cambodia had a strong incentive to stage exactly such a visit to rebut the exclusive-access narrative; and the berth the Cincinnati used has not been confirmed in open sources, if the ship used the older Cambodian pier rather than the Chinese-funded deep-draft pier, the observed division of berths is reinforced rather than weakened. The visit is therefore assessed as a Cambodian hedging signal consistent with the strategic calculus above, not as evidence that Chinese privileged access has lapsed.

These implications are contingent. They depend on access continuing, on the facilities being fitted out and garrisoned for sustained operations, and on the political durability of the Cambodia-China arrangement, none of which open sources can confirm. The judgment is therefore framed as a conditional projection rather than a description of an existing operational base.

Assessment: Continued privileged access would very likely enhance PLAN persistence and power-projection options toward the Malacca approaches and the southwestern South China Sea, while constraining allied access in the Gulf of Thailand. (Moderate confidence, contingent on the assumptions below.)

Indicators and Warning

Indicators that would raise confidence in privileged PLAN access: berthing of a PLAN combatant larger than a corvette, or a submarine, at the deep-draft pier; arrival of a submarine tender, replenishment ship, or dedicated PLAN logistics vessel; construction of a physical barrier or controlled checkpoint demarcating the northern sector; observable fuel, water, or ordnance throughput consistent with sustained operations; rotation patterns indicating PLAN crewing rather than RCN operation; and confirmation in any source of access-agreement terms.

Indicators that would lower confidence, favoring the open-facility or sovereignty-modernization hypotheses: a genuine, repeated multinational rotation through the deep-draft pier rather than one-off calls; confirmation that the resident corvettes are RCN-operated; removal or non-replacement of the resident corvettes without PLAN substitution; and Cambodian opening of the northern sector to routine third-party visits and documentation.

The January 2026 U.S. port call is a single data point on the disconfirming side; its weight depends on whether it is repeated and on which berth was used. A second high-value test will be whether China responds, diplomatically or by adjusting its presence, to the warming U.S.–Cambodia relationship.

Key Assumptions

- The reported infrastructure dimensions drawn from secondary commercial-imagery analysis are approximately accurate pending independent measurement. If the pier and dry-dock figures are materially wrong, the user-class judgments would weaken.

- The observed construction reflects intended military use absent contrary evidence. If significant portions are ultimately civilian or dual-use (e.g., the adjacent leased resort land), the basing judgment would soften.
- The sustained PLAN corvette presence reflects privileged access rather than an extended but ad hoc training engagement.
- The two resident corvettes are PLAN-operated rather than fully transferred to and crewed by the RCN. If they are RCN-operated, the access-pattern judgment weakens materially.
- The January 2026 U.S. port call reflects Cambodian hedging within a continuing Chinese-access arrangement rather than a genuine shift away from privileged Chinese access. If the U.S. opening proves durable and is repeated, the basing judgment softens.
- Open-source reporting from AMTI and major outlets accurately represents the imagery and events it describes; it is treated as reliable secondary corroboration, not primary collection.

Intelligence Gaps

- The terms of any Cambodia-China access agreement are unverified in open sources; reporting of a 30-year arrangement remains uncorroborated by primary documentation.
- The discrepancy in reported pier length (~363 m vs. ~650 m) is unresolved; the ~340 m measurement here favors the lower figure but should be confirmed against higher-resolution imagery.
- The intended user-class of the dry dock (submarine vs. corvette/RCN servicing) is undetermined and is the highest-value technical gap.
- The operational status of the two resident Type-056A corvettes (deployed PLAN units vs. transferred RCN hulls) is unresolved and is, with the agreement terms, the key discriminator between the privileged-access and sovereignty-modernization hypotheses.
- The berth used by the USS Cincinnati in January 2026 (deep-draft pier vs. older RCN pier) is unconfirmed and is directly relevant to the exclusivity judgment; it should be resolved against post-call imagery.
- Whether facilities are garrisoned, armed, and sensor-equipped for sustained operations cannot be determined from available optical imagery; higher-resolution or SAR-derived activity analysis is needed.

Source and Imagery Notes

This assessment uses free Copernicus Sentinel-2 and Sentinel-1 imagery, the SOAR Atlas archive, and Google and Apple Maps basemaps for historical baselines, supplemented by open reporting from the CSIS Asia Maritime Transparency Initiative, Radio Free Asia, Reuters, Nikkei Asia, the Lowy Institute, BlackSky public releases, and other press. Higher-resolution commercial imagery is cited through secondary reporting rather than first-party tasking. Imagery review was constrained by resolution limits, cloud cover, revisit gaps, and date misalignment between freely available scenes and the events of interest. All judgments rest on the alignment of imagery-observed change, infrastructure characteristics, and open reporting; no classified information was used.

Sources were weighted by reliability. The CSIS Asia Maritime Transparency Initiative and the Lowy Institute, both conducting independent imagery analysis or in-country fieldwork, were

treated as higher-reliability corroboration. Single-outlet press reports were treated as moderate-reliability and cross-checked where possible. Official statements from the Cambodian and Chinese governments were treated as interested-party claims requiring corroboration rather than as established fact.

Annex A. Construction and Activity Chronology

2020: Demolition of the former U.S.- and Australian-funded facilities; initial land clearing.

Jun 2022: Official groundbreaking on the Chinese-funded modernization.

Feb 2023: Satellite imagery shows two new piers under development (RFA).

2022–2023: Land clearing, dredging, first new building rows; dry-dock and slipway works begin.

Dec 2023: PLAN-pattern corvettes begin near-continuous presence at the new pier (AMTI).

Early 2024: AMTI tracks corvettes berthed around 93% of the period; first corvette rotation.

Nov 2024: Construction begins on new headquarters/administrative building.

2024: Clearing of reported air-defense HQ / naval radar site around 0.5 mi north of the base; activity later appears to stall.

Apr 2025: Hun Manet formally opens the expanded facilities, including the China-Cambodia Joint Support and Training Center.

2025: First foreign naval visits to the modernized base (Japanese vessels).

Jan 2026: USS Cincinnati becomes the first U.S. warship to dock at Ream post-renovation; INDOPACOM commander visits; Angkor Sentinel resumption and arms-embargo lift announced.